4. BEKTOREAK ESPAZIOAN

141 | 1)

a) $\bar{u}(1,-3,2)$ $\quad \bar{w}(5,-3,4)$

$\bar{v}(2,0,1)$ $\quad \bar{z}(-2,6,-4)$

$$A = \begin{pmatrix} 1 & -3 & 2 \\ 2 & 0 & 1 \\ 5 & -3 & 4 \\ -2 & 6 & -4 \end{pmatrix}$$

linealki indep bakoni hiru izan daiteke heina(A) ≤ 3

$$\left.\begin{array}{l} \begin{vmatrix} 1 & -3 \\ 2 & 0 \end{vmatrix} = 6 \neq 0 \\ \begin{vmatrix} 1 & -3 & 2 \\ 2 & 0 & 1 \\ 5 & -3 & 4 \end{vmatrix} = 0 \quad \begin{vmatrix} 1 & -3 & 2 \\ 2 & 0 & 1 \\ -2 & 6 & -4 \end{vmatrix} = 0. \end{array}\right\} \text{hein}(A) = 2$$

BI BEKTORE LINEALKI INDEPENT. DIRA

b) $\bar{w} = x\bar{u} + y \cdot \bar{v}$

$(5,-3,4) = x(1,-3,2) + y(2,0,1)$

$$\left\{\begin{array}{ll} x + 2y & = 5 \\ -3x & = -3 \\ \cancel{2x + y} & \cancel{= 4} \end{array}\right. \Rightarrow \left(\begin{array}{cc|c} 1 & 2 & 5 \\ -3 & 0 & -3 \\ 2 & 1 & 4 \end{array}\right)$$

$$\begin{vmatrix} 1 & 2 \\ -3 & 0 \end{vmatrix} = 6 \neq 0$$

$$\begin{vmatrix} 1 & 2 & 5 \\ -3 & 0 & -3 \\ 2 & 1 & 4 \end{vmatrix} = -12-15+24+3 = 0$$

hein(A) = hein(A') = 2

r = 3 $\quad$ r > 2

n = 2

(Ekuazio bat soberan)

$$\left\{\begin{array}{l} x + 2y = 5 \\ -3x = -3 \end{array}\right. \rightarrow \boxed{\begin{array}{l} y = 2 \\ x = 1 \end{array}}$$

$$\boxed{\bar{w} = \bar{u} + 2\bar{v}}$$

c) $\vec{w} = x(1,-3,2) + y(-2,6,-4)$
$(5,-3,4)$

$$\begin{cases} x-2y=5 \\ -3x+6y=-3 \\ 2x-4y=4 \end{cases} \qquad \left(\begin{array}{cc|c} 1 & -2 & 5 \\ -3 & 6 & -3 \\ 2 & -4 & 4 \end{array}\right)=A'$$

A (lehen bi zutabeak)

$A \rightarrow \begin{vmatrix} 1 & -2 \\ -3 & 6 \end{vmatrix} = 0$

$\begin{vmatrix} 1 & -2 \\ 2 & -4 \end{vmatrix} = 0$

$hein(A) = 1$

$A' \rightarrow \begin{vmatrix} -2 & 5 \\ 6 & -3 \end{vmatrix} = -24 \neq 0$

SISTEMA BATERAEZINA $\leftarrow$ $hein(A) \neq hein(A')$

Sistemak ez dauka soluziorik, beraz ezin da

d) $\bar{t}(-1,m,7)$, $\bar{u}$-ren eta $\bar{v}$-ren konb. lineala izateko

$\bar{t} = a\bar{u} + b\bar{v}$
$(-1,m,7) = a(1,-3,2) + b(2,0,1)$

$$\begin{cases} a+2b=-1 \\ -3a=m \\ 2a+b=7 \end{cases} \Rightarrow \left(\begin{array}{cc|c} 1 & 2 & -1 \\ -3 & 0 & m \\ 2 & 1 & 7 \end{array}\right)=A'$$

A (lehen bi zutabeak)

$\bar{t}$, beste kombinazio lineala izateko, sistemak soluzioa izan behar dau. (BATERAGARRIA)

Nola A 2×3 $\rightarrow$ $hein(A) = 2$ $\begin{vmatrix} 1 & 2 \\ -3 & 0 \end{vmatrix} = 6 \neq 0$.
beraz A', $hein(A') = 2$, bateragarria izateko.
Beraz konbinazio lineala izango da $hein(A') = 2$
$\rightarrow |A| = 0$ danean.

$$\begin{vmatrix} 1 & 2 & -1 \\ -3 & 0 & m \\ 2 & 1 & 7 \end{vmatrix} = 0$$

$4m + 3 + 42 - m = 0$

$3m - 45 = 0$

$\boxed{m = -15}$

$\boxed{\text{m=-15 danean, } \bar{t} \text{ bektorea } \bar{u} \text{ eta } \bar{v}\text{-ren konbinazio lineala da}}$

141) $\boxed{2}$ $\bar{x}(3,-1,0)$

$\bar{u}(1,2,-1)$

$\bar{v}(2,-3,5)$

Probatu behar da ea ea dala izan $\bar{x} = a\bar{u} + b\bar{v}$

$$(3,-1,0) = a(1,2,-1) + b(2,-3,5)$$

$$\begin{cases} a + 2b = 3 \\ 2a - 3b = -1 \\ -a + 5b = 0 \end{cases} \qquad \left(\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & -3 & -1 \\ -1 & 5 & 0 \end{array}\right)$$

kontrobatu behar doju sistemak ea daukolo soluziñoik.

$$A \rightarrow \begin{vmatrix} 1 & 2 \\ 2 & -3 \end{vmatrix} = -3-4 \neq -7 \qquad hein(A) = 2$$

$$A' \rightarrow \begin{vmatrix} 1 & 2 & 3 \\ 2 & -3 & -1 \\ -1 & 5 & 0 \end{vmatrix} = 2+30-9+5 = 28 \neq 0$$

$hein(A') = 3.$

SISTEMA BATERAEZIÑA da hein(A) ≠ hein(A') dalako

berat hein(A') = 3 dauez, 3 bektoreak linealki independienteak dira

4 Kalkulatu "m" eta "n" linealki menpekoak izateko.

a) $\bar{u}(m,-3,2)$
$\bar{v}(2,3,m)$
$\bar{w}(4,6,-4)$

$$\begin{vmatrix} m & -3 & 2 \\ 2 & 3 & m \\ 4 & 6 & -4 \end{vmatrix} = -\cancel{12m}-12m+\cancel{24}-24-\cancel{24}-6m^2 =$$
$$= -6m^2-24m-24$$
$$= -6(m^2+4m+4)$$
$$= -6(m+2)^2$$

Menpekoak izateko determ. berdin 0 izan behar da.

$$-6(m+2)^2=0 \longrightarrow \boxed{m=-2}$$

b) $\bar{u}(3,2,5)$
$\bar{v}(2,4,7)$
$\bar{w}(1,-1,n)$

$$\begin{vmatrix} 3 & 2 & 5 \\ 2 & 4 & 7 \\ 1 & -1 & n \end{vmatrix} = 8n+5$$

$8n+5=0$

$\boxed{n=-5/8}$ danean linealki menpekoak dira

5 OINARRI BAT IZATEKO: hiru bektoreak linealki independienteak izan behar dira. Ezin da izan bat, besteen konbinazio lineala. Beraz determinaltea $\neq 0$ izan behar da.

Adibidez:

a) $$\begin{vmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 2 & 2 & 2 \end{vmatrix} = 0+4+2-0-4-2 = \underline{\underline{0}}$$ Ezin da oinarria osatu.

c) $$\begin{vmatrix} -3 & 2 & 1 \\ 1 & 2 & -1 \\ 1 & 0 & 1 \end{vmatrix} = -6-2+0-2-2-3 = -15 \neq 0$$

OINARRIA da
linealki indep. direlako

BIDERKETA ESKALARRA

$\bar{u}\cdot\bar{v} = |\bar{u}|\cdot|\bar{v}|\cdot \cos\alpha.$

$\bar{u}\perp\bar{v} \longrightarrow \bar{u}\cdot\bar{v}=0.$

Oinarri ortogonala $\longrightarrow$ $\bar{u}\cdot\bar{v} = u_x\cdot v_x + u_y\cdot v_y + u_z\cdot v_z$

133) 2) $\bar{u}(5,-1,2)$
$\bar{v}(-1,2,-2)$

a) $\bar{u}\cdot\bar{v} =$

$\bar{u}\cdot\bar{v} = u_x v_x + u_y v_y + u_z v_z$

$\bar{u}\cdot\bar{v} = (5,-1,2)\cdot(-1,2,-2) = 5(-1)+(-1)\cdot 2 + 2(-2) =$

$= -5-2-4 = \underline{\underline{-11}}$

b) $|\bar{u}|$ eta $|\bar{v}|$

$|\bar{u}| = \sqrt{u_x^2+u_y^2+u_z^2}$

$|\bar{u}| = \sqrt{5^2+(-1)^2+2^2} = \sqrt{30}$

$|\bar{v}| = \sqrt{(-1)^2+2^2+(-2)^2} = \sqrt{9} = 3$ Modulua beti $\oplus$

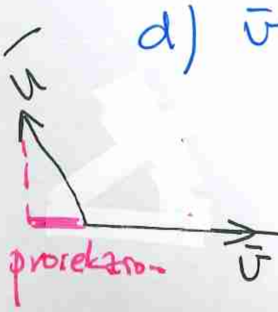
c) $(\widehat{\bar{u}\cdot\bar{v}}) = \alpha$

$$\cos\alpha = \frac{\bar{u}\cdot\bar{v}}{|\bar{u}|\cdot|\bar{v}|}$$

$$\cos\alpha = \frac{-11}{\sqrt{30}\cdot 3} = -0{,}669$$

$\arccos(-0{,}669) = 132^\circ\, 1'\, 26''$

d) $\bar{v}$-ren gaineko $\bar{u}$-ren proiekzioa



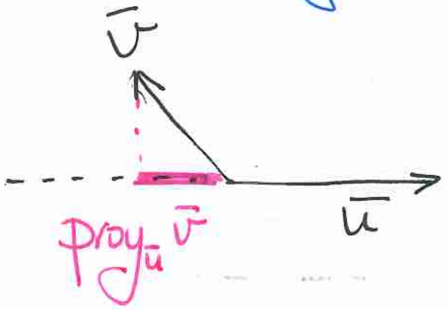
$$\text{proy}_{\bar{v}}\bar{u} = |\bar{u}|\cdot\cos\alpha = \cancel{|\bar{u}|}\cdot\frac{\bar{u}\cdot\bar{v}}{\cancel{|\bar{u}|}\,|\bar{v}|} = \frac{\bar{u}\cdot\bar{v}}{|\bar{v}|}$$

zubakia

$$\text{proy}_{\bar{v}}\bar{u} = \frac{-11}{3} = \boxed{-3{,}67}$$

Proiekzioaren modulua 3,67 eta $\bar{v}$-ren kontrako noranzkoa.

ū ren gaineko v̄-ren proiekzioa



$$proy_{\bar u}\bar v = |\bar v|\cdot\cos\alpha = \cancel{|\bar v|}\,\frac{\bar u\cdot\bar v}{|\bar u|\,\cancel{|\bar v|}}$$

$$proy_{\bar u}\bar v = \frac{\bar u\cdot\bar v}{|\bar u|} = \frac{-11}{\sqrt{30}} = \underline{\underline{-2{,}008}}$$

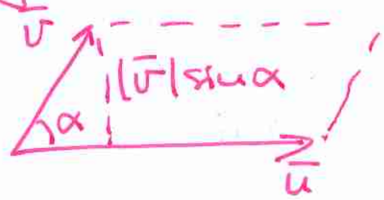
Proiekzioaren modulua 2,008 da eta ū-ren kontrako norantza.

BIDERKADURA BEKTORIALA.

- Bektore bat da (norabidea ū eta v̄ rekiko PERPENDIK

$$\boxed{\bar u\wedge\bar v = |\bar u|\cdot|\bar v|\cdot\sin\alpha.}$$

- ū eta v̄ norabide berdinekoak edo linealki menpekoak badira $\bar u\wedge\bar v = 0$.
- $\bar u\wedge\bar v$ ren MODULUA → paralelogramoaren AZALERA



- Beti betiko da $\bar u\wedge\bar u = 0$.

$$\bar u\wedge\bar v = \begin{vmatrix} \hat\imath & \hat\jmath & \hat k \\ u_x & u_y & u_z \\ v_x & v_y & v_z \end{vmatrix}.$$

7. $\bar a(1,2,2)$
$\bar b(-4,5,-3)$

e) $\bar a\wedge\bar b = (1,2,2)\cdot\wedge(-4,5,-3) =$

$$= \begin{vmatrix} \hat\imath & \hat\jmath & \hat k \\ 1 & 2 & 2 \\ -4 & +5 & -3 \end{vmatrix} = -6\hat\imath - 8\hat\jmath + 5\hat k + 8\hat k + 3\hat\jmath - 10\hat\imath$$

$$= -16\hat\imath - 5\hat\jmath + 13\hat k$$

$$\boxed{\bar a\wedge\bar b = (-16,-5,13)}$$

a) $\bar{a}\cdot\bar{b} =$

Bid. eskalarra $\bar{u}\cdot\bar{v} = u_x\cdot v_x + u_y v_y + u_z v_z$

$\bar{a}\cdot\bar{b} = (1,2,2)\cdot(-4,5,-3) =$
$= 1(-4) + 2\cdot 5 + 2(-3) = 0 \rightarrow$ berat $\underline{\underline{\bar{a}\perp\bar{b}}}$

b) $|\bar{a}|$ eta $|\bar{b}|$ $\quad |\bar{u}| = \sqrt{u_x^2 + u_y^2 + u_z^2}$

$|\bar{a}| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{9} = 3$

$|\bar{b}| = \sqrt{(-4)^2 + 5^2 + (-3)^2} = \sqrt{50} = 5\sqrt{2}.$

c) $(\widehat{\bar{a}\cdot\bar{b}}) = \alpha.$

$\cos\alpha = \dfrac{\bar{u}\cdot\bar{v}}{|\bar{u}|\cdot|\bar{v}|} = \dfrac{0}{3\,5\sqrt{2}} = 0 \longrightarrow \alpha = 90^0 \quad \underline{\underline{\bar{a}\perp\bar{b}}}$

d) $(\bar{a}+\bar{b})(\bar{a}-\bar{b}) = (\bar{a})^2 - (\bar{b})^2$

$(\bar{a})^2 = \bar{a}\cdot\bar{a} = |\bar{a}|\cdot|\bar{a}|\cdot\cos 0 = |\bar{a}|^2$

$(\bar{a}+\bar{b})\cdot(\bar{a}-\bar{b}) = |\bar{a}|^2 - |\bar{b}|^2 = 3^2 - (\sqrt{50})^2 = \underline{\underline{-41}}.$

e) $\bar{a}\wedge\bar{b} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & 2 & 2 \\ -4 & 5 & -3 \end{vmatrix} =$

$= -6\hat{\imath} - 8\hat{\jmath} + 5\hat{k} + 8\hat{k} + 3\hat{\jmath} - 10\hat{\imath} =$

$= -16\hat{\imath} - 5\hat{\jmath} + 13\hat{k} \longrightarrow \underline{\underline{(-16,-5,13)}}$

f) $|\bar{a}\wedge\bar{b}| = \sqrt{(-16)^2 + (-5)^2 + 13^2} = 15\sqrt{2}$

g) $\bar{b}$-ren gaineko $\bar{a}$-ren proiek $= \text{proy}_{\bar{b}}\bar{a}$

$\text{proy}_{\bar{b}}\bar{a} = |\bar{a}|\cdot\cos\alpha = \cancel{|\bar{a}|}\cdot\dfrac{\bar{a}\cdot\bar{b}}{\cancel{|\bar{a}|}\,|\bar{b}|} = \dfrac{\bar{a}\cdot\bar{b}}{|\bar{b}|} = \underline{\underline{0.}}$

h) $\bar{a}$-ren gaineko $\bar{b}$-ren proiek $= \text{proy}_{\bar{a}}\bar{b} = \underline{\underline{0}}$

$\underline{\underline{\bar{a}\perp\bar{b}}}$

8 $\bar{a} = \hat{i} + m\hat{j} + \hat{k} = (1, m, 1)$

$\bar{b} = -2\hat{i} + 4\hat{j} + m\hat{k} = (-2, 4, m)$

a) m paraleloak izateko.

PARALELOAK IZATEKO

$\bar{u}$ eta $\bar{v}$ norabide berdinekoak (proportzionalak) edo linealki menpekoak izango dira eta beraz $\bar{u} \wedge \bar{v} = 0$.

$$\frac{1}{-2} = \frac{m}{4} = \frac{1}{m} \longrightarrow \boxed{m = -2}$$

$$0 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & m & 1 \\ -2 & 4 & m \end{vmatrix} = m^2\hat{i} - 2\hat{j} + 4\hat{k} + 2m\hat{k} - m\hat{j} - 4\hat{i} = 0$$

$$\left| \begin{array}{ll} m^2 - 4 = 0 & m = \pm 2 \\ -2 - m = 0 & m = -2 \\ 4 + 2m = 0 & m = -2 \end{array} \right\} \Rightarrow \boxed{m = -2}$$

b) m ORTOGONALAK IZATEKO.

ORTOGONALAK / PERPENDIKULARRAK izateko.

$\bar{u} \cdot \bar{v} = 0 \longrightarrow \bar{u} \perp \bar{v}$

$$\bar{a} \cdot \bar{b} = (1, m, 1) \cdot (-2, 4, m) = -2 + 4m + m = 0.$$

$$5m - 2 = 0$$

$$\boxed{m = 2/5}$$

9

10 $\bar{a}(1,2,3)$ $\bar{b}(2,-2,1)$

Ortogonalak izateko $\bar{a}\cdot\bar{b}=0$

$(1,2,3)\cdot(2,-2,1) = 2-4+3 = 1 \neq 0$ Ez dira ortogonalak

$$\cos\alpha = \frac{\bar{a}\cdot\bar{b}}{|\bar{a}||\bar{b}|} = \frac{1}{\sqrt{1^2+2^2+3^2}\cdot\sqrt{2^2+2^2+1^2}} = \frac{1}{\sqrt{14}\cdot 3} =$$

$= 0{,}089 \longrightarrow \boxed{\alpha = 84^\circ\ 53'\ 20''}$

11 m para ser ortogonal.

$\bar{a}(1,3,m)$ $\bar{b}(1,-2,3)$

$\bar{a}\bar{b} = (1,3,m)(1,-2,3) = 1-6+3m = -5+3m$

$\bar{a}\cdot\bar{b}=0 \longrightarrow -5+3m=0$ $\boxed{m=5/3}$

12 $\bar{u}=\left(\frac{1}{2}, \frac{1}{2}, 0\right)$

UNITARIO IZATEKO $\boxed{|\bar{u}|=1}$

$$|\bar{u}| = \sqrt{\left(\frac{1}{2}\right)^2+\left(\frac{1}{2}\right)^2+0^2} = \sqrt{\frac{2}{4}} = \frac{\sqrt{2}}{2} \neq 1$$ Ez da unitarioa

BEKTORE UNITARIOA, NORANTZA BARDINEKOA $\boxed{\frac{\bar{u}}{|\bar{u}|}}$

$$\frac{\bar{u}}{|\bar{u}|} = \frac{(1/2,\ 1/2,\ 0)}{\sqrt{2}/2} =$$

$\frac{1}{2} : \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

$$\left(\frac{1}{2}:\frac{\sqrt{2}}{2},\ \frac{1}{2}:\frac{\sqrt{2}}{2},\ 0\right) = \left(\frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}},\ 0\right) = \left(\frac{\sqrt{2}}{2},\ \frac{\sqrt{2}}{2},\ 0\right)$$

Bi bektore posible dagoz: $\boxed{\begin{array}{l}\left(\frac{\sqrt{2}}{2},\ \frac{\sqrt{2}}{2},\ 0\right)\\ \left(-\frac{\sqrt{2}}{2},\ -\frac{\sqrt{2}}{2},\ 0\right)\end{array}}$

13 —

14 PARALELOGRAMOAREN AZALERA = $\bar{u}\wedge\bar{v}$-ren MODULUA

$$\hat{u}\wedge\bar{v} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & k \\ 2 & -1 & 1 \\ -1 & 3 & 2 \end{vmatrix} = -2\hat{\imath} - \hat{\jmath} + 6\hat{k} - \hat{k} - 4\hat{\jmath} - 3\hat{\imath}$$

$$= -5\hat{\imath} - 5\hat{\jmath} + 5\hat{k}$$

$$\hat{u}\wedge\bar{v} = (-5,-5,5)$$

$$|\bar{u}\wedge\bar{v}| = \sqrt{(-5)^2+(-5)^2+5^2} = \sqrt{75} = 5\sqrt{3} = \underline{\underline{8{,}66}}$$

$$\boxed{A = 8{,}66\ u^2}$$

15 $\bar{u}(2,3,1)$ eta $\bar{v}(-1,3,0)$ rekiko $\perp$ den bektore unitarioa

· kalkulatuko dugu $\bar{u}$ eta $\bar{v}$ rekiko perpendikularra den $\bar{w}$

$$\bar{w} = \bar{u}\wedge\bar{v} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & 3 & 1 \\ -1 & 3 & 0 \end{vmatrix} = (-3,-1,9)$$

· Bektore unitarioa

$$\frac{\bar{w}}{|\bar{w}|} = \frac{(-3,-1,9)}{\sqrt{91}} = \boxed{\left(\frac{-3}{\sqrt{91}}, \frac{-1}{\sqrt{91}}, \frac{9}{\sqrt{91}}\right)}$$

$$|\bar{w}| = \sqrt{(-3)^2+(-1)^2+9^2} = \sqrt{91}$$

16 $\bar{u}(1,-1,0)$ $\bar{v}(2,0,1)$

$\bar{w}\perp\bar{u}$, $\bar{w}\perp\bar{v}$

$|\bar{w}| = \sqrt{24}$

$$\bar{w} = \bar{u}\wedge\bar{v} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & -1 & 0 \\ 2 & 0 & 1 \end{vmatrix} = -\hat{\imath} - \hat{\jmath} + 2\hat{k}$$

unitarioa $\dfrac{(-1,-1,2)}{\sqrt{1+1+2^2}} = \dfrac{1}{\sqrt{6}}(-1,-1,2)$

Modulua $\sqrt{24}$ izatiko:

$$\frac{\sqrt{24}}{\sqrt{6}}(-1,-1,2) = \frac{2\cancel{\sqrt{6}}}{\cancel{\sqrt{6}}}(-1,-1,2) = 2(-1,-1,2)$$

$$= (-2,-2,4)$$

Bilatzen popatzoz bektorea $\underline{\underline{(-2,-2,4)}}$ edo $(2,2,-4)$

BIDERKADURA MISTOA

$$\bar{u}\cdot(\bar{v}\wedge\bar{w}) = \begin{vmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix} = \text{ZENBAKI BAT DA}$$

(paralelepipedoaren BOLUMENA)

$\bar{u}\cdot(\bar{v}\wedge\bar{w}) = 0 \rightarrow$ planokideak

$\boxed{17}$ a) $\bar{u}(1,-3,2)$ $\bar{v}(1,0,-1)$, $\bar{w}(2,3,0)$

$$\bar{u}\cdot(\bar{v}\wedge\bar{w}) = \begin{vmatrix} 1 & -3 & 2 \\ 1 & 0 & -1 \\ 2 & 3 & 0 \end{vmatrix} = 15 \leftarrow \frac{\text{Bolumena}}{\underline{\underline{15u^3}}}$$

b) ... $| \quad | = -15 \rightarrow$ Bolumena $15u^3$

c) ... $| \quad | = 0 \rightarrow$ Bektoreak planokideak

18 —

19 $\bar{a}(3,-1,1)$ $\bar{b}(1,7,2)$ $\bar{c}(2,1,-4)$

$$[\bar{a},\bar{b},\bar{c}] = \bar{a}(\bar{b}\wedge\bar{c}) = \begin{vmatrix} 3 & -1 & 1 \\ 1 & 7 & 2 \\ 2 & 1 & -4 \end{vmatrix} = -111.$$

Bolumena $= \boxed{111\ u^3}$

Tetraedroa $\rightarrow$ $\boxed{\frac{1}{6}\,111 = 18{,}5\ u^3}$

Tetraedroaren Bolumena $\boxed{\frac{1}{6}\,\bar{u}(\bar{v}\wedge\bar{w})}$

20 PLANOKIDEAK $\boxed{\bar{u}\cdot(\bar{v}\wedge\bar{w}) = 0}$

$$\begin{vmatrix} 2 & -3 & 1 \\ 1 & m & 3 \\ -4 & 5 & -1 \end{vmatrix} = 2m+8$$

$2m+8=0$

$\boxed{m=-4}$